

# Ashwini Koppula

Data Analyst

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## Summary

Data Analyst with expertise in financial and clinical data analysis, statistical modeling, and data visualization. Proficient in SQL, Python (Pandas, NumPy, SciPy, PySpark), R, Power BI, and Tableau to extract insights, optimize workflows, and drive data-driven decision-making. Experienced in cloud platforms (AWS S3, AWS Glue, Azure SQL) and ETL tools, ensuring efficient data integration and transformation. Adept at leveraging advanced BI dashboards and automation to enhance business performance.

## Technical Skills

- Programming & Scripting:** Python (Pandas, NumPy, SciPy, PySpark), R (dplyr, tidyr, caret, stats, forecast, glmnet), SQL, DAX
- Data Analysis & Modeling:** Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Predictive Modeling, Time-Series Forecasting
- Data Visualization & Reporting:** Power BI (Bookmarks, Drillthrough, Dynamic Parameters), Tableau (LOD Expressions, R Integration), Matplotlib, Seaborn, ggplot2, plotly
- Database Management:** Azure SQL Database, SQL Server, AWS S3
- Cloud & ETL Tools:** AWS Glue, Azure Data Factory
- Business Intelligence & Automation:** Power BI, Tableau, Excel, R Markdown, RServe, JIRA, GitHub, Agile Methodology

## Professional Experience

### Data Analyst, LPL Financial

10/2024 – Present | Remote, USA

- Worked on Revenue and Cost Analysis for Financial Products (e.g., mutual funds and retirement plans) and collaborated with cross-functional teams using Agile methodology, optimizing the cost-to-revenue ratio by 15% for a portfolio of \$500M in assets.
- Utilized advanced SQL techniques such as CTEs and Window Functions to extract, aggregate, and manipulate over 2 million rows of financial data, enabling accurate reporting on product performance and increasing data retrieval efficiency by 25%.
- Conducted high-level analysis of financial product performance, assessing revenue trends, cost fluctuations, and profitability metrics for a diverse range of products, leading to a 10% improvement in cost efficiency and a 5% increase in overall revenue.
- Employed Python with libraries such as Pandas for data manipulation, NumPy for numerical analysis, SciPy for statistical computations, and PySpark for processing large datasets (up to 1TB), optimizing analysis time by 40% for large-scale reports.
- Leveraged AWS S3 for storing large financial datasets and used AWS Glue for ETL processes, streamlining data preparation, reducing processing time by 30%, and ensuring seamless integration of multiple data sources for reporting.
- Designed high-level, interactive Power BI dashboards incorporating advanced features like Bookmarks, Drillthrough, and Dynamic Parameters to visualize real-time revenue and cost insights, empowering executives to track product performance and optimize strategies for a portfolio valued at \$2B.

### Associate Data Analyst, Novartis

05/2021 – 07/2023 | Hyderabad, India

- Performed Clinical Trial Data Analysis by cleaning and transforming over 1 million rows of raw clinical trial data using Python (Pandas, NumPy) and R (dplyr, tidyr), improving data accuracy by 25% and ensuring high-quality insights for drug trials.
- Leveraged R (caret, stats) for statistical modeling, hypothesis testing, and survival analysis, enabling in-depth evaluation of clinical trial efficacy and reducing decision uncertainty by 20%.
- Utilized SQL (Azure SQL Database) for complex data extraction and transformation, applying advanced SQL techniques (window functions, CTEs, and pivoting) to optimize query performance, reducing data retrieval time by 30%.
- Conducted advanced EDA using Python (Pandas, Matplotlib, Seaborn) and R (ggplot2, plotly) to detect patterns, correlations, and outliers, uncovering hidden trends that improved clinical trial monitoring accuracy by 15%.
- Integrated Python and R for multivariate analysis and predictive modeling, using Python (SciPy, NumPy) for data processing and R (forecast, glmnet) for time-series forecasting and regression models, improving patient outcome predictions.
- Managed clinical trial data across Azure SQL Database, integrating with Excel and R Markdown for dynamic reporting, reducing manual data preparation efforts by 40%.
- Automated ETL pipelines with Azure Data Factory and R scripts, streamlining data ingestion and transformation, increasing workflow efficiency by 35%.
- Developed interactive Tableau dashboards with LOD expressions, R integrations (via RServe), and calculated fields to visualize clinical trial progress, improving decision-making speed by 20%.

## Education

### Master of Science in Data Science

University of Maryland, Baltimore County (UMBC)

2023 – 2025 | Maryland, USA

### Bachelor of Technology in Electrical and Electronics Engineering

Malla Reddy College of Engineering and Technology

2018 – 2022 | Hyderabad, India

## Certifications

AWS Certified Cloud Practitioner (2025)

ITIL Foundation Certificate in IT service management (2023)

